SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 2 **Duration : 45 Minutes**
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
   Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Question 1: Intelligent Agents in Smart Home Automation System.

Intro:-

A smart home automation system uses AI to control the lighting, temperature, and security based on sensors inputs and user preferences. The four types of intelligent agents are simple reflex, model-based, goal-based, utility-based agents.

1. Simple Reflex Agent:-

A simple reflex agent takes action based on the current input using predefined rules.

Examples:-

* If the motion is detected, switch on the light
* If temperature is high, turn on the AC
* If an intruder is detected, activate the alarm

Advantages:- Fast, simple and easy to implement

2. Model-Based Agent:-

A model based agent maintains an internal state using current and previous observations.

Examples:-

* Remembers which rooms are occupied
* Tracks temperature and lighting status
* Uses previous observations to control lights.

Advantages:-

Handles incomplete information and makes better decisions.

Disadvantages:-

Requires memory and an accurate internal model.

3. Goal-Based Agent:-

A goal-based agent selects actions to achieve the specific objectives.

Examples:-

* Maintains the desired room temperature
* Secures the house when the user leaves
* Turn off unnecessary lights to save energy.

Advantages:- supports planning and goal-oriented decisions.

4. Utility-Based Agent:-

A utility based agent selects actions that provides the greatest overall benefit by balancing comfort, energy consumption, and security.

Examples:- Adjusts the AC temperature to maintain comfort while reducing electricity usage.

Advantages:- Balances multiple objectives and compares the alternative actions.

Comparision:-

Agent	Main function
Simple Reflex	Reacts to current inputs
Model-based	Maintains internal state
Goal-based	Achieved specific goals
Utility-based	Balances multiple objectives

Conclusion:-

A hybrid agent provides fast responses, informed decisions, goal-oriented planning, and efficient resource planning.

Question 2: Solving an Unknown Maze Using UCS and IDS

Intro:-

A robot in an unknown maze must find the exit without prior knowledge of the layout. Uniform cost search (UCS) and Iterative Deepening search (IDS) are uniformed search algorithms that can help find a path.

1. Uniform Cost Search (UCS):-

UCS expands the node with the lowest cummulative path cost.

Steps:-

① Add the starting node to a priority queue with cost zero
② Expand the node with the lowest cost
③ check wheather it is the goal
④ Add neighbouring nodes with their cummulative costs.
⑤ Repeat until the exit is found

Advantages:-

* Finds the minimum-cost path when step costs are non-negative under standard UCS assumptions.

2. Iterative Deepening Seach (IDS)

IDS repeteadly performs depth-limited DFS, increasing the depth limit each time.

Steps:-

① Start with depth limit zero
② Search up to the current depth-limit
③ Increase the limit if the goal is not found
④ Repeat until the exit is reached

3. Time and space complexity

Let b be the branching factor and d the depth of the shallowest goal

Feature	UCS	IDS
Time complexity	Exponential in the worst case	$O(b^d)$
Space complexity	Exponential in the worst case	$O(bd)$
optimality	yes, under standard cost assumptions	yes, for equal-cost steps
Main advantage	Minimum-cost path	Low memory usage
Main disadvantage	High memory usage	Repeated exploration

4. Most-suitable combination:

A model-based, goal-based agent combined with online exploration and UCS is suitable when the robot must minimize travel cost.

If all moves have equal cost and memory is limited, IDS is a reasonable alternative.

Conclusion:-

UCS is preferable for minimum-cost navigation, while IDS is useful for memory-efficient search with equal-cost moves.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand